

# Erdős Problem #1061

EP-1061

L1 - TRACTABLE

OPEN

Number Theory

## ORIGINAL PROBLEM STATEMENT

How many solutions are there to  $\sigma(a) + \sigma(b) = \sigma(a + b)$  with  $a + b \leq x$ , where  $\sigma$  is the sum of divisors function? Is it  $\sim cx$  for some constant  $c > 0$ ?

**Source and attribution.** Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.  
Archive: <https://www.unsignedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>